

Virtual machine (VM): This is the special environment that VirtualBox creates for your guest OS while it is running. All its resources, like the CPU, memory, storage, network devices, etc, are virtual.

Guest additions. These are additional software bundles installed inside a guest to improve its performance and extend its functionality. For instance, these allow us to share folders between the host and guest, and have the drag-and-drop functionality.

Installation

VirtualBox is freely available for personal and commercial use. Visit <https://www.virtualbox.org/wiki/Downloads> and download it for the Windows platform. At the time of writing this article, the latest version of VirtualBox was 6.0.12.

VirtualBox installation on Windows is pretty straightforward, just like any other software. After downloading the installer, just double-click on it and follow the on-screen instructions. If everything goes fine, you should see the window of the VirtualBox manager, as shown in Figure 1.

Basic operations

Now our lab environment is ready. Let us perform some primitive VirtualBox operations like creating a VM, saving it, powering it off, shutting it down and, finally, deleting it.

Creating a VM: A virtual machine is an emulation of a computer system. Just like any other physical system, it has memory, a CPU, a network, storage and other peripheral devices. To create the virtual machine, perform the following tasks in sequence:

- Launch VirtualBox manager
- Navigate to *Machine->New menu*
- Select the name and operating system
- Provide memory for the virtual machine
- Select the 'Create a virtual hard disk

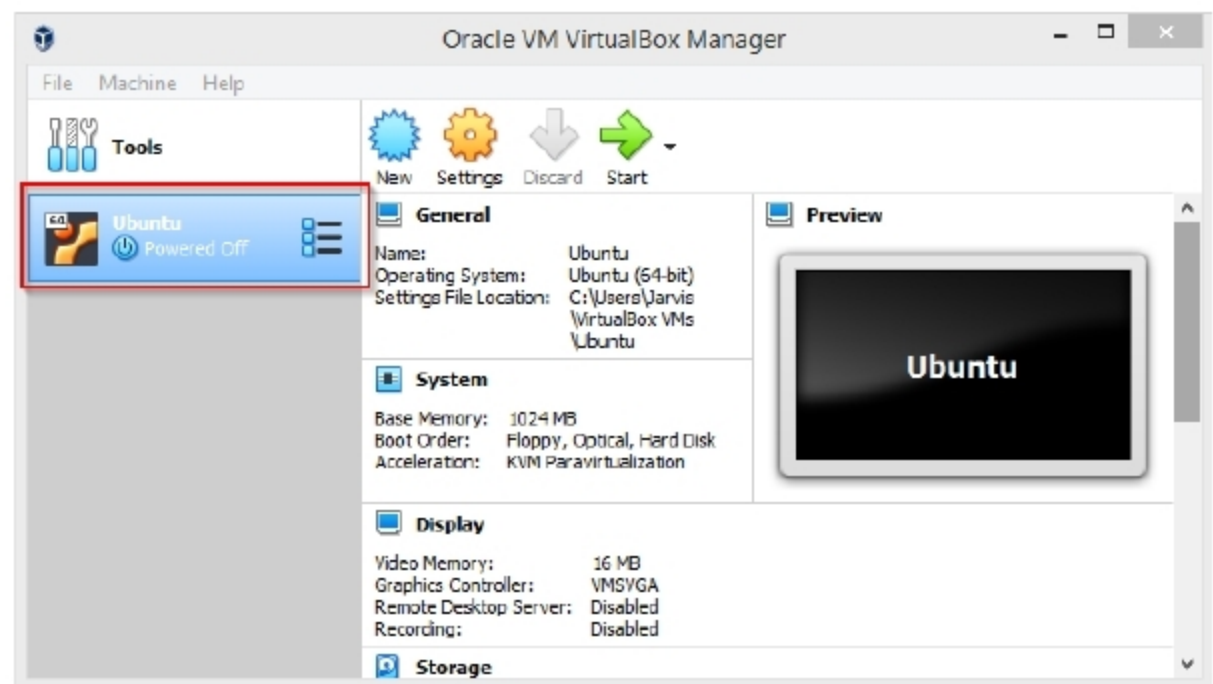


Figure 2: Creating a VM

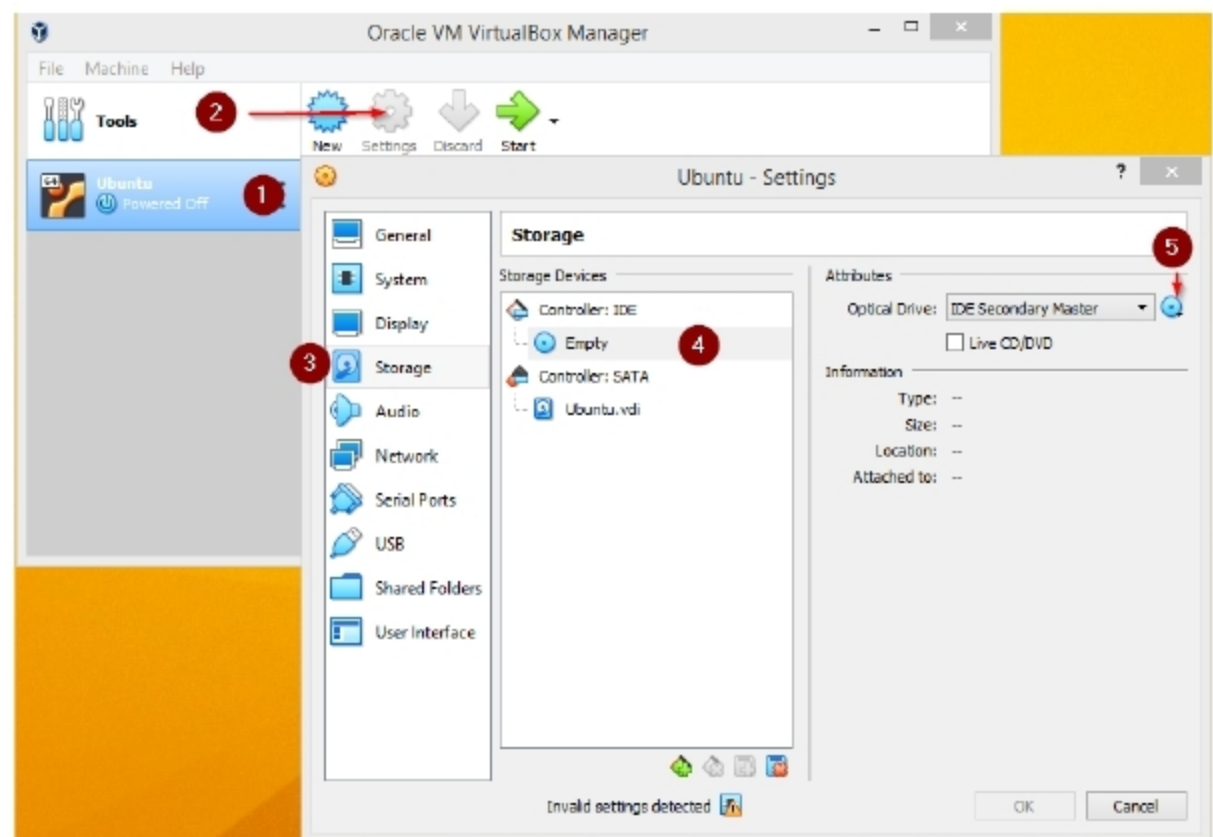


Figure 3: Installing the guest OS

- now' option, as a brand new virtual machine is being created
- Select VDI as the hard disk file type
- Select 'Dynamically allocated' as the storage option
- Select the file location and size
- Click on the *Create* button

After performing these operations, a new VM will appear in VirtualBox manager as shown in Figure 2.

Installing the guest OS: To begin installing the OS, we need to attach an ISO image to the VM. Follow the steps given below to install a guest OS:

- Select the newly created VM from

- the previous sequence
- Click the *Settings* button on the toolbar
- Select the storage option from the left pane
- Select the optical disk drive from the storage devices
- Provide the path of the ISO image and click the *OK* button. Figure 3 depicts the first five steps.
- Select the VM from the left pane. Click the 'Start' button on the toolbar. Follow the on-screen instructions to complete OS installation